Machine Learning MCQs (1–50)

1.

What is the primary goal of Machine Learning?

A) To replace databases
B) To enable systems to learn from data and improve automatically
C) To increase hardware speed
D) To create websites

2.

Which of the following is a type of supervised learning?

A) Clustering
B) PCA
C) Classification
D) Association Rule Mining

3.

Which of the following is an example of unsupervised learning?

A) Linear Regression
B) Logistic Regression
C) K-Means Clustering
D) Decision Tree

4.

In supervised learning, the training data contains:

A) Only inputs
B) Only outputs
C) Inputs and corresponding outputs
D) No labels

5.

Regression is mainly used to predict:

A) Categories
B) Continuous values
C) Clusters
D) Rules

6.

Classification is used to predict:

A) Numerical values
B) Continuous outputs
C) Discrete class labels
D) Eigenvectors

7.

Overfitting occurs when a model:

A) Performs poorly on training data
B) Learns noise in training data
C) Has too few features
D) Uses too little data

8.

Underfitting occurs when a model:

A) Is too complex
B) Learns noise
C) Cannot capture the underlying pattern
D) Has very high variance

9.

A cost function is used to:

A) Measure prediction error
B) Store training data
C) Reduce dimensions
D) Calculate clusters

10.

Which dataset is used to evaluate model performance after training?

A) Training set
B) Validation set
C) Test set
D) Feature set

11.

Linear Regression is primarily used for:

A) Classification
B) Clustering
C) Regression
D) Reinforcement Learning

12.

The relationship in simple linear regression is represented by:

A) $y = mx + c$
B) $y = x^2$
C) $y = \log(x)$
D) $y = \sin(x)$

13.

Which assumption is important in Linear Regression?

A) Features should be independent of output
B) Linear relationship between variables
C) Data must be categorical
D) No training data required

14.

Logistic Regression is mainly used for:

A) Clustering
B) Regression only
C) Classification
D) Dimensionality Reduction

15.

The output range of the sigmoid function is:

A) $(-\infty, \infty)$
B) [0,1]
C) [-1,1]
D) [0,100]

16.

The sigmoid function is also called:

A) Activation Function
B) Logistic Function

C) Cost Function
D) Kernel Function

17.

Why is Logistic Regression considered a classification algorithm?

A) It predicts probabilities and classes
B) It uses clusters
C) It predicts continuous values
D) It reduces dimensions

18.

Which algorithm is known as a Lazy Learner?

A) SVM
B) KNN
C) Naïve Bayes
D) PCA

19.

KNN stands for:

A) K-Normal Network
B) K-Nearest Node
C) K-Nearest Neighbor
D) Kernel Neural Network

20.

KNN is called non-parametric because:

A) It assumes no fixed data distribution
B) It uses parameters only
C) It uses regression
D) It requires labels

21.

Why is an odd value of K generally preferred in KNN?

A) Faster computation
B) Reduces tie situations
C) More clusters formed
D) Better visualization

22.

The distance most commonly used in KNN is:

A) Hamming Distance
B) Manhattan Distance
C) Euclidean Distance
D) Jaccard Distance

23.

KNN can be used for:

A) Classification only
B) Regression only
C) Both Classification and Regression
D) Clustering only

24.

Naïve Bayes is based on:

A) Bayes Theorem
B) Pythagoras Theorem
C) Gradient Descent
D) Eigenvalues

25.

The main assumption in Naïve Bayes is:

A) Features are dependent
B) Features are independent
C) Data is clustered
D) Data is sparse

26.

Conditional Probability refers to:

A) Probability of A given B
B) Probability of A and B together only
C) Probability of neither A nor B
D) None of these

27.

Decision Trees split data based on:

A) Information Gain or Gini Index
B) PCA
C) K value
D) Odds Ratio

28.

Information Gain is used to measure:

A) Accuracy
B) Entropy Reduction
C) Distance
D) Precision

29.

The root node of a Decision Tree represents:

A) Final class
B) First splitting attribute
C) Leaf node
D) Test set

30.

SVM stands for:

A) Support Vector Machine
B) Supervised Variable Model
C) Statistical Vector Method
D) Simple Variance Model

31.

The main objective of SVM is to:

A) Minimize clusters
B) Find the optimal separating hyperplane
C) Reduce dimensions
D) Calculate probabilities

32.

Support Vectors are:

A) Cluster centers
B) Data points closest to decision boundary

C) Test samples
D) Eigenvectors

33.

Which technique enables SVM to handle non-linear data?

A) Clustering
B) Sigmoid Function
C) Kernel Trick
D) PCA

34.

A confusion matrix is used to evaluate:

A) Clustering
B) Classification Models
C) PCA
D) Matrix Factorization

35.

True Positive (TP) means:

A) Positive predicted as positive
B) Negative predicted as positive
C) Positive predicted as negative
D) Negative predicted as negative

36.

Precision is calculated as:

A) TP/(TP+FN)
B) TP/(TP+FP)
C) TN/(TN+FP)
D) FP/(TP+FP)

37.

Recall is calculated as:

A) TP/(TP+FN)
B) TP/(TP+FP)
C) TN/(TN+FN)
D) FP/(FP+TN)

38.

Accuracy of a classifier is:

A) Correct Predictions / Total Predictions
B) TP / FP
C) FP / Total Predictions
D) FN / Total Predictions

39.

Clustering is an example of:

A) Supervised Learning
B) Reinforcement Learning
C) Unsupervised Learning
D) Semi-supervised Learning

40.

The goal of clustering is to:

A) Predict labels
B) Group similar data points
C) Minimize cost function
D) Perform regression

41.

K-Means clustering requires specifying:

A) Number of Features
B) Number of Classes
C) Number of Clusters (K)
D) Number of Labels

42.

The center of a K-Means cluster is called:

A) Support Vector
B) Hyperplane
C) Centroid
D) Eigenvector

43.

The Elbow Method is used to determine:

A) Learning Rate
B) Optimal K value
C) Accuracy
D) Entropy

44.

Which algorithm is more sensitive to outliers?

A) K-Medoids
B) K-Means
C) Naïve Bayes
D) PCA

45.

In K-Medoids, cluster centers are:

A) Mean values
B) Random points
C) Actual data points
D) Eigenvectors

46.

PCA stands for:

A) Principal Component Analysis
B) Primary Cluster Analysis
C) Probability Component Algorithm
D) Principal Classification Analysis

47.

The main purpose of PCA is:

A) Classification
B) Regression
C) Dimensionality Reduction
D) Clustering

48.

Before applying PCA, data is usually:

A) Deleted
B) Standardized
C) Clustered
D) Classified

49.

PCA transforms correlated features into:

A) Missing values
B) Clusters
C) Uncorrelated Principal Components
D) Labels

50.

Matrix Factorization is widely used in:

A) Recommendation Systems
B) Decision Trees
C) Logistic Regression
D) SVM